Numericals Problems: unit 4

1. Show by direct substitution into the time dependent Schrodinger equation for the free particle, that $\psi(x,t)=A\cos(kx-\omega t)$ is not a solution.
2. For a free quantum particle show that the wave function $\psi(x,t)=A\cos kx\, e^{-i\omega t}$ satisfies the time dependent Schrodinger equation.
3. Explain, why the following Eigen function are not acceptable solution of the Schrodinger equation.
   a. $\chi(x)=0$ for x< 0
   = A Cos kx for x ≥ 0
   b. $\chi(x)=A\dfrac{e^{ikx}}{x}$ c. $\chi(x)=A\ln kx$
4. What is the probability of finding a particle in a well of width ‘a’ at a position a/4 from the wall if n=1, if n=2, if n=3. Use the normalized wave function, $\psi(x,t)=\sqrt{\dfrac{2}{a}}\sin\dfrac{n\pi x}{a}e^{\frac{-iEt}{\hbar}}$.

   Ans P_1 = 1/a ; P_2 = 2/a ; P_3 = 1 /a
5. Consider a particle of mass ‘m’ which can move freely along x-axis. The wave function for the particle in the lowest energy state is given by0z

   $\psi(x,t)=A\cos\dfrac{\pi x}{a}e^{\frac{-iEt}{\hbar}}$ for $\dfrac{-a}{2}$ < x > $\dfrac{a}{2}$

   = 0 for x ≤ $\dfrac{-a}{2}$ and x ≥ $\dfrac{a}{2}$

   Where, A is arbitrary real constant and E is the total energy of particle verifying that $\psi(x,t)$ is the solution of Schrodinger wave equation. Determine the value of E.

   Ans $E=\dfrac{\hbar^2\pi^2}{2ma^2}$
6. Normalized the one dimensional wave function given by

   $\psi(x,t)=A\sin\dfrac{\pi x}{a}$ for 0 < x < a

   = 0 for outside Ans A= $\dfrac{2}{a}$ and $\psi(x,t)=\sqrt{\dfrac{2}{a}}\sin\dfrac{\pi x}{a}$
7. Wave function of the particle confined in a box of length L is $\psi(x,t)=\sqrt{\dfrac{2}{L}}\sin\dfrac{\pi x}{L}$. Calculate the probability of finding the probability of finding the particle in the region 0 < x < $\dfrac{L}{2}$ and $\dfrac{-L}{2}$ < x < $\dfrac{L}{2}$.

   Ans P=1/2 and P= 1
8. Given a normalized wave function $\psi(x)=\sqrt{\alpha}e-\alpha x$. Find the probability of existence of the particle between x = $\dfrac{1}{\alpha}$ and x = $\dfrac{2}{\alpha}$.

   Ans P= $\dfrac{e^{-2}-e^{4}}{2}$
9. Find the expectation value for momentum and energy of particles defined by its matter wave equation $\psi(x,t)=Ae^{i(kx-\omega t)}$.

   Ans $\langle p\rangle=\hbar k$; $\langle E\rangle=\hbar\omega$
10. Consider the particle in the ground state is represented by a wave function $\psi(x)=B\sin\dfrac{\pi x}{a}$ within 0< x < a, where $B=\sqrt{\dfrac{2}{a}}$. What is a) the average position b) the average momentum and c) the average energy of such particle ?

    Ans $\langle x\rangle=\dfrac{a}{2}$ $\langle p\rangle=0$ $\langle E\rangle=E_0$

11. a. How many atomic states are there in Hydrogen with n=3 ?
    b. How are they distributed among the sub shells ? Label each state with appropriate set of Quantum numbers n, l, m_l, m_s.
    c. Show that the number of states in a shell, that is, states having the same n, is given by $2n^2$.
12. Calculate the normal Zeeman splitting of the calcium 4226 A^0 line when the atoms are placed in a magnetic field of 1.2 Tesla. Ans: 0.0996 A^0.
13. In a normal Zeeman experiment, the spectral line of wavelength 500 nm split into three lines separated by 0.28 A^0 in a magnetic field of 3T. Calculate the value of the specific charge of electron from these data.
14. The calcium line of wavelength 4226 A^0 exhibits normal Zeeman splitting when placed in a uniform magnetic field of 4T. Calculate the wavelength of three components of normal Zeeman pattern and the separation between them.
15. A beam of Hydrogen atom is used in Stern-Gerlach type experiment. The atom emerges from the oven with a velocity 10^4 m/sec. They enter a region 20 cm long where there is a magnetic field gradient $3X10^4$ T/m. The field gradient is perpendicular to the incident velocity of the atoms. The mass of the Hydrogen atom is 1.67 X 10^{-27} kg. What is the separation of the two components of the beam as they emerge from the magnet. Ans: 3.3 cm
16. In Stern-Gerlach experiment, a beam of silver atoms (atomic weight 108) enters as inhomogenous magnetic field perpendicular to it with a velocity of 10^4 m/sec. If length of field is 10 cm and gradient of field is 10^2 J/m. Calculate, the total reflection of beam on photography plate assuming that the magnetic moment of silver atoms is only due to spin motion of electron. Ans 5.7 X 10^{-7} m.